

Cognitive-structured Multimodal Agent for Multimodal Understanding, Generation, and Editing

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Abstract

Recent unified multimodal models demonstrate that a single architecture can jointly perform vision/language understanding and image generation/editing. However, these monolithic designs rely on repeatedly feeding all historical visual and textual inputs into a shared context window, limiting scalability in long-horizon multimodal dialogue due to visual token explosion and unreliable cross-turn visual referencing. In this work, we propose a Cognitive-structured Multimodal Agent that externalizes visual information into an Episodic Visual Memory and selectively reactivates relevant visual episodes during reasoning. The agent consists of a Perceptual Abstraction Engine for structured visual abstraction, a Cognitive Retrieval Engine for cross-turn memory retrieval, and a Multimodal Executive Controller for autonomous task inference and action planning. To address the lack of turn-level retrieval supervision in existing multimodal dialogue datasets, we further develop a Unified Scenario Engine that programmatically generates structured multi-turn conversations with fine-grained retrieval annotations, enabling reinforcement learning to optimize perceptual abstraction and retrieval policies. We additionally construct a long-horizon visual-dialogue benchmark and stratify it by difficulty to evaluate episodic visual recall. Extensive experiments show that our 8B agent achieves 91.4% retrieval accuracy over 20-turn sessions, surpassing 32B baselines by +8.2%, while nearly halving per-turn inference time (23.1s $\rightarrow$ 12.7s). We further present the Cognitive-structured Multimodal Agent Harness (CMA-Harness), a tool-augmented deployment of the same cognitive structure that integrates persistent multimodal memory, web access, image generation/editing/composition tools, and OpenAI-compatible serving. These results suggest that structured memory and modular decision-making provide a more scalable and efficient paradigm for long-horizon multimodal agents than monolithic parameter scaling. Our code and dataset will be open-sourced on GitHub.

1. Introduction

Unified multimodal models have recently shown that a single large architecture can jointly perform vision/language understanding and image generation. Approaches [6, 7, 12, 22, 34, 39] consolidate perception and generation within a shared parameter space and achieve strong performance across diverse tasks. These systems typically treat multimodal interaction as autoregressive prediction over a unified token sequence.

While effective for short-context interaction, this unified paradigm exhibits structural limitations in long-horizon multimodal dialogue. In practical image-text conversations, users frequently reference images introduced many turns earlier, request iterative modifications, or switch between understanding and generation without explicit task indicators. Under such settings, repeatedly injecting all historical visual tokens into the context window leads to two major issues. First, visual tokens are substantially more expensive than text tokens; as dialogue length increases, token usage grows rapidly and reduces effective reasoning capacity. Second, reliance on implicit attention over extended contexts weakens cross-turn visual referencing, resulting in retrieval errors and semantic drift. These challenges indicate that parameter scaling alone is insufficient for sustained multimodal interaction.

Recent agent-based approaches attempt to address parts of this problem. Memory-augmented video agents [8, 42] maintain episodic buffers for long video reasoning, and multi-agent generation systems [30, 36] coordinate specialized modules for image creation and editing. However, these methods either target a single modality or a single task, and none jointly addresses cross-turn visual retrieval, mixed-task orchestration, and long-horizon context management within a unified dialogue framework.

Building a multimodal agent capable of long-horizon dialogue requires explicit memory management and autonomous decision-making. Instead of treating images as persistent tokens, we argue that visual information should be externalized into structured memory units and selectively activated when necessary.

*Work done during an internship at Tencent WeChat Team.

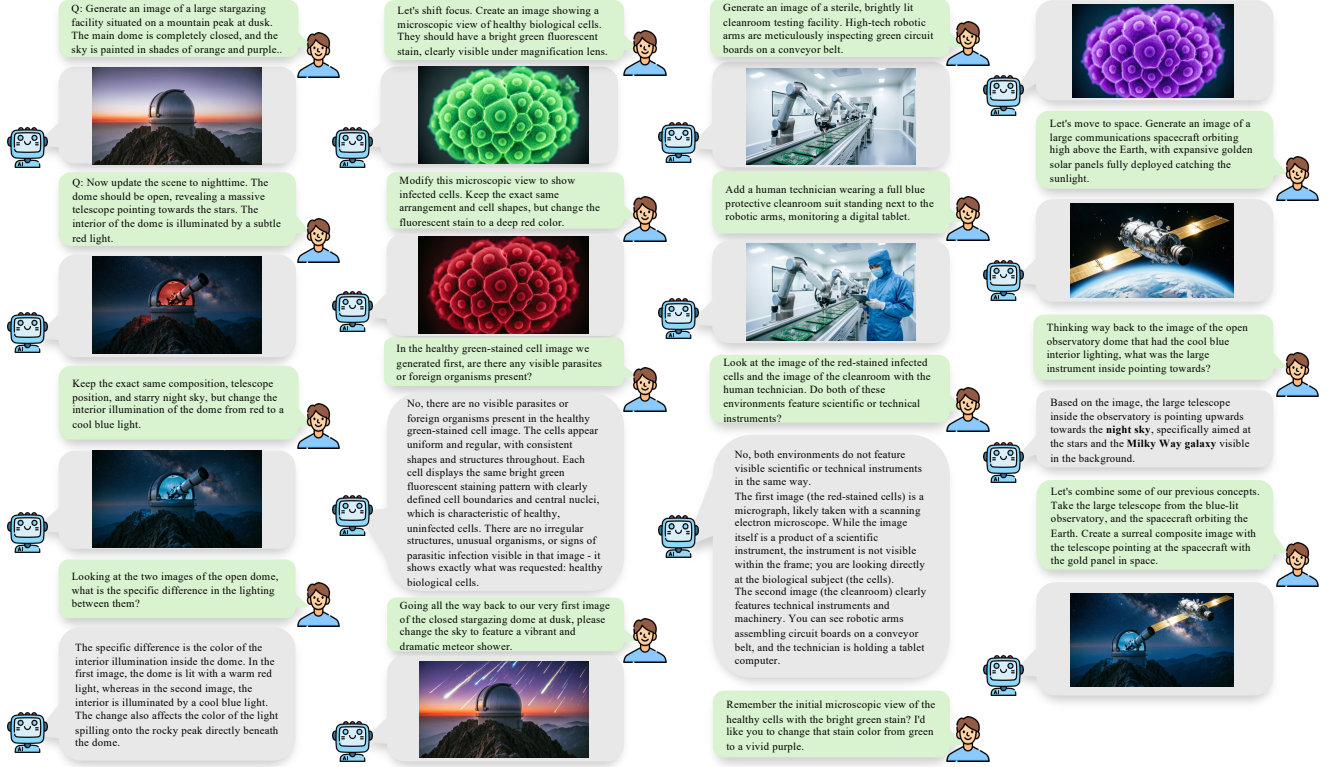


Figure 1. A multi-turn multimodal dialogue produced by our Cognitive-structured Multimodal Agent, spanning long turns across multiple topics (stargazing dome, biological cells, cleanroom facility, spacecraft). The agent autonomously handles interleaved understanding, generation, and editing tasks while accurately retrieving and referencing visual episodes from earlier turns.

To this end, as Fig. 2 shows, we propose a **Cognitive-structured Multimodal Agent** that decouples perception, episodic memory, retrieval reasoning, and executive control. Incoming images are first processed by a **Perceptual Abstraction Engine (PAE)**, which produces structured semantic abstractions including descriptive captions, attribute tags, and compact images. These representations are stored in an external **Episodic Visual Memory (EVM)**, allowing visual content to persist without occupying the dialogue context window. Given the evolving dialogue state, a **Cognitive Retrieval Engine (CoRE)** performs cross-modal reasoning to retrieve relevant visual episodes. An **Multimodal Executive Controller (MEC)** integrates dialogue context and retrieved memory, autonomously infers task intent (understanding, retrieval, generation, or editing), and plans the final action.

By retrieving only relevant visual episodes, the agent avoids repeatedly feeding all historical images into the model, significantly reducing visual token overhead while maintaining long-range consistency. Furthermore, the decoupling of the execution engine (MEC) from the retrieval engine (CoRE) enables independent advancements for each module, enhancing the versatility and capabilities of the over-

all agent. This architectural design allows for the integration with models of varying scales, which helps in achieving an optimal trade-off between performance and inference latency.

However, developing such memory-aware agents introduces a new challenge: existing multimodal dialogue datasets rarely provide explicit supervision for cross-turn visual retrieval. Most datasets focus on single-turn grounding or short-context reasoning, and therefore lack fine-grained annotations indicating which historical visual episode should be recalled at each dialogue turn. To address this limitation, we introduce a **Unified Scenario Engine** that programmatically generates structured multi-turn multimodal conversations. The engine simulates diverse visual scenes and dialogue trajectories, automatically producing fine-grained turn-level retrieval annotations that specify which visual episode is relevant to each query. Using these data, we further apply reinforcement learning to jointly optimize perceptual abstraction and retrieval strategies: the Perceptual Abstraction Engine is trained to produce representations that maximize downstream retrieval success, while the Cognitive Retrieval Engine learns retrieval policies guided by correctness-based reward signals.

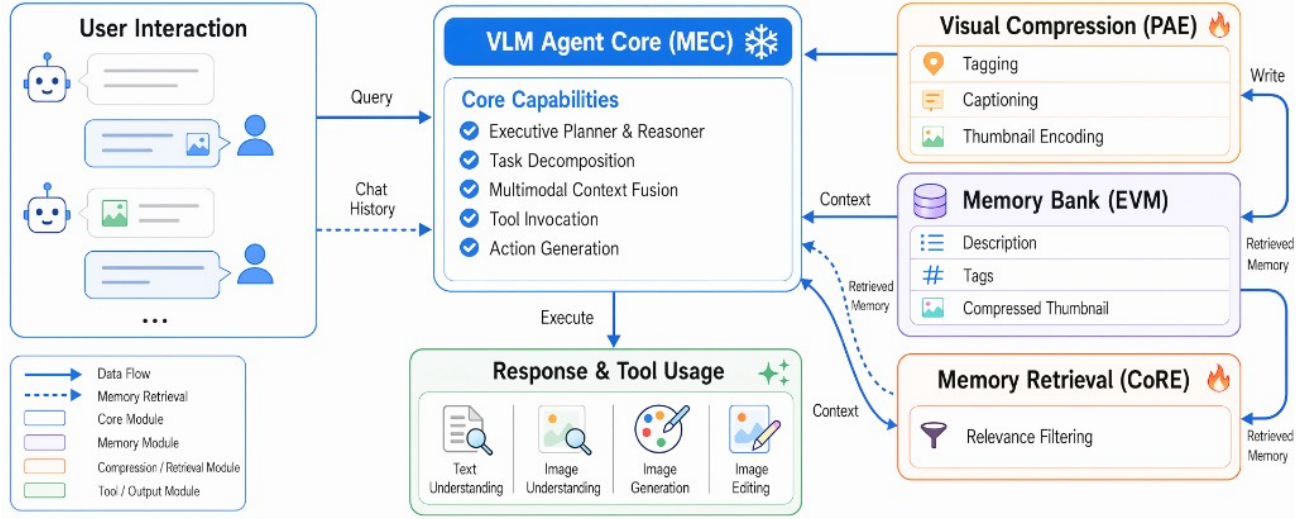


Figure 2. Pipeline of our proposed Cognitive-structured Multimodal Agent.

To systematically evaluate long-horizon visual memory and agent behavior, we additionally construct a **Multi-turn Context Agent Benchmark**, consisting of 100 dialogues, each with 20 rounds, and each dialogue randomly interleaved with multiple tasks. The benchmark stratifies retrieval difficulty by temporal distance between the query and referenced image (four difficulty levels defined based on retrieval horizon length, the frequency of topic shifts, and the proportion of multi-image retrieval), enabling controlled evaluation of cross-turn visual recall. Unlike existing multimodal benchmarks that focus on single-turn reasoning, this benchmark specifically targets episodic memory retrieval in sustained multimodal interaction.

Extensive experiments demonstrate that our 8B agent consistently outperforms unified 32B baselines on long-range image retrieval accuracy and multi-turn visual consistency. The advantage becomes increasingly pronounced as dialogue horizons grow, while our approach maintains substantially lower visual token usage. Beyond the benchmark-optimized agent, we further instantiate the same cognitive structure as the Cognitive-structured Multimodal Agent Harness (CMA-Harness), which combines persistent multimodal memory with web search, image generation, editing, composition, and OpenAI-compatible serving. These results indicate that structured memory and modular decision-making provide a more scalable foundation for multimodal agents than monolithic parameter scaling. Figure 1 presents visualization of our agent, demonstrating its ability to effectively leverage contextual information across dialogue turns.

To summarize, our main contributions are as follows: (1) We propose a Cognitive-structured Multimodal Agent with explicit episodic visual memory for long-horizon dialogue

that enables autonomous task inference. (2) We develop a Unified Scenario Engine that programmatically generates structured multimodal conversations with turn-level retrieval supervision. (3) We construct a benchmark specifically designed to evaluate cross-turn visual retrieval in multimodal interaction. (4) We introduce reinforcement learning-based optimization techniques for perceptual abstraction and retrieval, enabling an 8B agent to surpass unified 32B baselines in long-range multimodal dialogue. (5) We present CMA-Harness, a tool-augmented deployment of the same architecture that extends the memory-centric agent to open-ended multimodal workflows.

2. Related Work

2.1. Agents

LLM & VLM Agents. Large Language Models (LLMs) [1, 2, 29, 40] and Vision-Language Models (VLMs) [4, 17, 19, 24, 33, 45] have driven the development of autonomous agents capable of deliberative reasoning and tool execution, often utilizing paradigms like reasoning traces or step-wise preference alignment [18, 26]. To address context window limitations, recent LLM and VLM agents incorporate explicit memory mechanisms [14, 23, 41]. By externalizing information into episodic and semantic memory banks [13, 42], these agents mitigate the prohibitive cost of attending to long conversational histories while significantly improving retrieval accuracy over extended interactions. In multimodal domains, VLM agents [15, 20, 21, 31] frequently act as controllers to orchestrate visual tools for specific, long-form tasks such as video understanding [8, 35].

Single & Multi-agents. While single-agent systems [9, 36] excel at routing sub-tasks to expert tools with self-correction,

multi-agent frameworks [10, 16, 25, 30] distribute complex goals—such as creative generation—across specialized roles using programmable, collaborative protocols. However, existing visual agents typically operate within short contexts or restrict their scope to isolated domains. Crucially, our work introduces the first long-horizon, multi-agent architecture that seamlessly unifies pure text dialogue, multimodal understanding, image generation, and image editing within a single conversational framework. By leveraging an external episodic visual memory and a decoupled retrieval engine, our system maintains cross-turn intent and visual consistency without the exponential token costs of standard context scaling.

2.2. Multimodal Models

Foundational Vision-Language Models (VLMs) such as Qwen-VL [3] family models have rapidly progressed from contrastive representation learning to instruction-following assistants capable of complex reasoning. Concurrently, unified multimodal models [7, 27] aim to consolidate perception and generation into a shared parameter space. Token-based autoregressive models [6, 7, 34] interleave text and discretized visual tokens into a single sequence. Meanwhile, hybrid architectures [22, 39] blend autoregression with continuous diffusion or rectified flow to optimize both understanding and generative quality. In parallel, text-guided image generation and editing are heavily driven by diffusion architectures; frameworks like InstructPix2Pix [5] and TalkPhoto [11] enable instruction-driven manipulation, with recent methods explicitly addressing error accumulation in multi-turn iterative editing [44].

Despite the elegance of unified foundation models, scaling the context window is insufficient for long-horizon interactions. Benchmarks such as MMNeedle [32] and Visual Haystacks [38] reveal that relying on implicit attention over massive visual histories leads to severe retrieval degradation and semantic drift. Furthermore, repeatedly encoding visual tokens in unified architectures is computationally exorbitant. By externalizing visual history into structured episodes, our framework bypasses the inherent token limits of unified models, effectively maintaining reliable, long-range visual grounding across interleaved understanding, generation, and editing tasks.

3. Methods

3.1. Problem Formulation

We formalize long-horizon multimodal dialogue as an interaction trajectory:

$$\mathcal{D} = \{(u_t, x_t, y_t)\}_{t=1}^T,$$

where u_t denotes the user text input at turn t , x_t the optional visual input, and y_t the system response. The dialogue

history before turn t is $\mathcal{H}_{<t} = \{(u_i, x_i, y_i)\}_{i=1}^{t-1}$, and the historical image set is $\mathcal{I}_{<t} = \{x_i\}_{i=1}^{t-1}$.

A naive approach feeds the full history into a multimodal model:

$$y_t = f(u_t, x_t, \mathcal{H}_{<t}),$$

which suffers from three limitations as t grows: (i) prohibitive visual-token growth, (ii) contextual interference from irrelevant images, and (iii) unstable reasoning over long histories.

While text history carries essential semantic continuity, only a small *subset* of past images is relevant at each turn—a fundamental asymmetry overlooked by existing unified models. We reformulate the problem as **visual-context selection**, learning a policy π that selects a minimal relevant image subset:

$$\mathcal{I}_t^* = \pi(u_t, \mathcal{H}_{<t}), \quad \mathcal{I}_t^* \subseteq \mathcal{I}_{<t},$$

with the system prediction

$$y_t = f(u_t, \text{Text}(\mathcal{H}_{<t}), \mathcal{I}_t^*).$$

Our objective is to learn π that minimizes $|\mathcal{I}_t^*|$ while preserving downstream task performance. The Cognitive-structured Multimodal Agent framework (Sec. 3.2) implements π via coordinated specialized agents.

3.2. Cognitive-structured Multimodal Agent

Relying on a single monolithic model to compress memory, select context, and execute tasks results in conflicting objectives and unbounded visual-token growth. This limitation is confirmed empirically by near-zero retrieval accuracy of unified baselines (Sec. 5.3).

We therefore decompose the pipeline into the **Cognitive-structured Multimodal Agent** framework (Fig. 2), which assigns each sub-problem to a dedicated specialist, making every component independently trainable and upgradable:

- **Perceptual Abstraction Engine (PAE)**: abstracts each incoming image into a structured memory entry, including semantic tags, textual description, and a compressed thumbnail, which is stored in the EVM.
- **Cognitive Retrieval Engine (CoRE)**: retrieves indices of task-relevant visual episodes from EVM given the current query.
- **Multimodal Executive Controller (MEC)**: performs task classification (generate, edit, understand, or pure chat), orchestrates PAE and CoRE accordingly, and produces the final system output.

Orchestration loop. Here, an *orchestration loop* denotes a single interaction iteration with the language model (i.e., one dialogue turn with ChatGPT). Within each orchestration loop, MEC executes the following procedure:

1. **Task identification.** MEC first classifies the incoming user request to determine the task type (e.g., *generation/editing* versus *understanding/chat*).
2. **Memory retrieval and context assembly.** For tasks that require visual context, MEC queries the visual-memory module (CoRE) to retrieve relevant memory entries. The retrieved visual entries are then fused with the dialogue history and the current user instructions to form a unified context representation.
3. **Task-specific dispatch.**
 - (a) *Generation / editing:* MEC rewrites the unified context into a refined, model-ready prompt for the image-generation/execution model (e.g., Qwen-Image-Edit [37]).
 - (b) *Understanding / chat:* MEC directly generates a text response based on the unified context.

To limit computational cost, only the retrieved subset of historical images is forwarded to the executor in each orchestration loop. This bounds the number of visual tokens processed per loop and yields approximately a $2\times$ inference speedup compared to all-context baselines (see Sec. 5.5).

3.3. Optimizing memory construction and retrieval

Accurate retrieval of relevant historical images is critical for downstream performance. We jointly optimize PAE (memory construction) and CoRE (memory retrieval) for retrieval success, rather than standalone captioning quality. Training data is exclusively drawn from the Unified Scenario Engine pool (Sec. 3.4), while the evaluation benchmark M2CA-Bench is strictly held out.

3.3.1. Supervised Fine-Tuning of Retriever

Each SFT sample pairs memory entries (tags, descriptions, thumbnails) with a user query; the target is the ground-truth retrieval set. We use 300 dialogue sessions (20 turns each), split 90:10 for training and validation.

3.3.2. Reinforcement Learning for Retriever (DAPO)

We apply DAPO [43] to further optimize retrieval policies. The reward is the difficulty-weighted Jaccard similarity between predicted and ground-truth retrieval sets:

$$r = \frac{|\hat{R} \cap R^*|}{|\hat{R} \cup R^*|} \cdot w_d,$$

where $\hat{R}$ is the prediction, R^* the ground truth, and $w_d = 1.2$ for *hard/very_hard* turns (1.0 otherwise). Soft length penalties discourage malformed outputs. RL samples are filtered to emphasize (i) incorrect baseline predictions, (ii) *hard/very_hard* turns, and (iii) negative samples requiring no retrieval, mitigating shortcut learning.

Table 1. Retrieval difficulty taxonomy.

Difficulty	Characteristic Patterns	Typical Tags
easy	Same-topic sequential generation or editing.	—
medium	High-similarity variants; simple topic switch.	topic_switch
hard	Cross-topic retrieval; long-range callback (≥ 8 turns).	cross_topic long_term_mem
very_hard	Multi-image comparison; fusion edits; ambiguous references.	cross_topic long_term_mem

3.3.3. Reinforcement Learning for Memory Writer (DAPO)

Rather than optimizing traditional captioning metrics (CIDEr, BERTScore), we directly reward memory entries based on retrieval effectiveness. Specifically, CoRE is frozen, and retrieval accuracy on memory entries produced by PAE serves as the reward:

$$r = \text{Jaccard}(\hat{R}, R^*) - \mathcal{L}_{\text{format}}.$$

$\mathcal{L}_{\text{format}}$ represents the format punishment. This closes the loop, ensuring memory representations are optimized for end-task utility.

3.4. Training Data & Benchmark Construction

Existing long-horizon multimodal dialogue datasets lack turn-level retrieval supervision. We therefore build a Unified Scenario Engine that programmatically generates structured 20-turn sessions with dense retrieval annotations (Fig. 3).

The pipeline operates as a closed loop: (1) A human-simulation module (Gemini) samples a topic, generates a query, its ground-truth retrieval set, and a difficulty tag.; (2) A zero-shot multi-agent system (Qwen-VL + Qwen-Image-Edit) produces a candidate answer; (3) A *GT-justify* mechanism verifies retrieval correctness; mismatches are corrected; (4) The verified answer is fed back to Gemini to generate the next turn.

Table 2 shows the data splits of the training data and M2CA-Bench. Training and evaluation sessions are produced with disjoint random seeds and non-overlapping topics; we denote the held-out evaluation set as **Multi-turn Context Agent Benchmark (M2CA-Bench)**.

3.4.1. Scenario Representation and Topic Ontology

Each session is structured as:

$$S = \{(t_i, \tau_i, R_i^*, d_i, \mathbf{f}_i)\}_{i=1}^T,$$

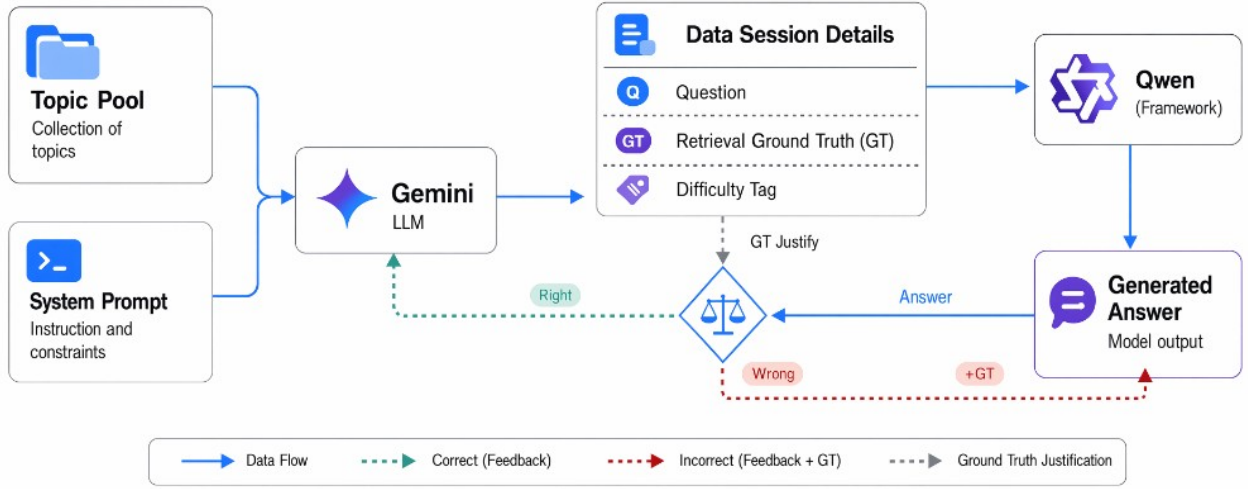


Figure 3. Pipeline for data generation, validation, and split construction in the Unified Scenario Engine.

Table 2. Data splits derived from the Unified Scenario Engine.

Split	Scale	Turns	Usage
SFT train	270 sessions $\times$ 20 turns	$\sim$ 5,400	Retrieval SFT
SFT val	30 sessions $\times$ 20 turns	$\sim$ 600	Validation
RL	Filtered hard subset	varies	DAPO training
M2CA-Bench	100 sessions $\times$ 20 turns	2,000	Evaluation

where t_i is the user input, τ_i the task type, R_i^* the ground-truth retrieval set, d_i the difficulty, and f_i encodes retrieval challenge tags.

We curate 55 topics across eight domains (commercial, industrial, educational, public service, hospitality, natural landscape, scientific, and space); each topic defines *generate*, *edit*, *cross-reference-edit*, and *understand* prompts to ensure compositionality. Gemini [28] synthesizes dialogue flows with structural inductive biases (e.g., “at least 3 long-range callbacks with span ≥ 8 turns”), ensuring compositional and controllable conversations.

3.4.2. Retrieval Difficulty & Hard-Negative Design

Turns are stratified into four difficulty levels based on topic shift, temporal span, multi-image interaction, and ambiguity (Table 1).

To prevent shortcut learning, we include: (i) high-similarity confounders (near-duplicate images differing subtly), and (ii) negative retrieval samples (semantic negatives requiring no retrieval, and structural negatives prompting new generation).

3.4.3. Held-Out Benchmark

The held-out benchmark consists of 100 sessions $\times$ 20 turns, generated with disjoint seeds and topics. Annotation fidelity

is ensured via LLM validation and manual review. Evaluation metrics are detailed in Sec. 5.2.

3.5. Implementation Details

Structured memory format. For every incoming or generated image, PAE writes a structured memory entry containing a compact thumbnail, semantic tags, and a natural-language description. The memory-writing prompt requires valid JSON with fields such as *tags* and *description*; this format makes the stored episode both machine-readable and interpretable. In practice, tags provide high-recall lexical anchors, while the description preserves spatial, relational, and stylistic cues that are difficult to capture with short labels alone.

Multimodal retrieval interface. CoRE receives a list of candidate thumbnails aligned with candidate indices, textual memory entries containing tags and descriptions, and the current user query. It outputs an image-index list, with an empty list indicating that no historical image is needed. This interface is shared by SFT, RL, and evaluation, ensuring that the model is optimized under the same input-output contract used at inference time.

Executive controller modes. MEC classifies each turn into generation, editing, composition, understanding, or pure-chat modes. For generation, it rewrites short user requests into detailed visual prompts. For editing and composition, it grounds the instruction in retrieved images and emits a precise editing prompt that specifies the target region or object while preserving unrelated content. For understanding and chat, it answers directly using the retrieved visual evidence, dialogue history, and memory metadata.

Generation and editing backbones. Text-to-image generation is handled by Qwen-Image, while image editing

and multi-image composition use Qwen-Image-Edit. Unless otherwise stated, inference uses 50 denoising steps and a classifier-free guidance scale of 4.0. During data construction, we use a faster Qwen-Image-Edit-2511-Lightning configuration with four denoising steps to reduce generation cost; final evaluation uses the standard high-quality pipeline. **Prompt interfaces.** The module prompts are intentionally format-constrained so that memory writing, retrieval, task classification, prompt rewriting, and visual answering expose stable input-output contracts across SFT, RL, evaluation, and deployment. We provide the full prompt templates in Appendix A.

4. CMA-Harness: A Tool-Augmented Engineering Deployment Framework

We introduce the Cognitive-structured Multimodal Agent Harness (CMA-Harness), a tool-augmented deployment of the same cognitive-structured architecture in Fig. 2. The goal is not to replace the Cognitive-structured Multimodal Agent framework with a separate system, but to extend its response-and-tool-usage branch while preserving the original division of labor among PAE, EVM, CoRE, and MEC. In the benchmark setting, this decomposition already works well when the user’s intent is grounded in dialogue-local visual history and no external facts or references are needed. Practical workflows, however, often require information beyond the current session, such as real product images, logos, names, news events, screenshots, or web pages. They also benefit from remembering user-specific preferences and project context accumulated across conversations, which makes the agent feel more consistent and helps its outputs better match the user’s actual needs. CMA-Harness therefore keeps Fig. 2’s memory-centric structure, assigns tool selection and invocation to MEC as the executive controller, expands the response module with external and deterministic tools, and strengthens the memory layer for persistent multi-session interaction.

4.1. From Core Agent System to Deployed Harness

CMA-Harness follows the same information flow as Cognitive-structured Multimodal Agent. User interactions first enter a dialogue state that contains the current instruction, uploaded images, and prior turns. New visual inputs are first persisted as memory assets and then asynchronously abstracted by a PAE-like card extractor into structured tags, captions, palettes, and thumbnails. These entries are written into EVM and later retrieved by CoRE when the user refers to previous visual context. The central difference lies in the operational scope of MEC. In the original benchmark pipeline, MEC mainly decides whether the task requires understanding, retrieval, generation, or editing, and then prepares the prompt for the downstream multimodal model. In CMA-Harness, MEC becomes a tool-calling executive: it

still performs task decomposition and multimodal context fusion, but it can additionally call external and deterministic tools before producing the final answer.

This design keeps the cognitive interpretation of Fig. 2. PAE remains responsible for visual compression, including tags, captions, and thumbnails. EVM remains the long-horizon visual memory substrate. CoRE remains the retrieval module that selects relevant memory for the current turn. MEC remains the main reasoning module, but its action space is enlarged from a small set of generation/editing choices to a structured set of callable tools. As a result, CMA-Harness is best understood as a harness around Cognitive-structured Multimodal Agent: it provides service interfaces, tool schemas, persistent storage, and execution policies around the same cognitive modules.

4.2. MEC-Driven Action Space Expansion

CMA-Harness exposes a registry of 17 tools to MEC. Each tool is described by a name, natural-language capability description, JSON schema, concurrency-safety flag, progress message, and asynchronous execution function. The registry is converted into an OpenAI-style function-calling interface, so MEC can reason about the user’s goal, decide whether tools are necessary, choose the appropriate tool, fill its arguments, observe the result, and continue the loop if another action is needed. The tools extend the original response-and-tool-usage block in Fig. 2: generation and editing tools include image generation, image editing, multi-image composition, deterministic collage construction, background removal, watermark removal, cropping, and text overlay; understanding and memory tools include image inspection, best-image selection, and image retrieval; external-information tools include web search, batch search, web-image search, web-page fetching, and image fetching.

The key design choice is that tool access is governed by an MEC action policy rather than exposed as an undifferentiated toolbox. MEC first decides whether a tool is needed at all; many requests are better answered directly without external execution. When the requested output is a purely imagined scene, MEC can call text-to-image generation. When the output must contain a specific real-world visual entity, such as a logo, product, screenshot, artwork, or named person’s face, MEC is instructed not to rely on a verbal description, because text-only generation or editing will often hallucinate the identity. Instead, it first obtains real pixels through memory retrieval or web-image fetching, and then routes the task to composition or editing with those image identifiers as visual references.

CMA-Harness also separates semantic composition from deterministic layout. Multi-image diffusion composition is used when several visual references must be fused into a new scene. In contrast, poster, infographic, timeline, comparison, and many-item showcase tasks often require faithful faces,

logos, captions, dates, or CJK text. For these cases, MEC can choose a PIL-based collage tool that preserves each input by pixel-level placement and renders text deterministically, optionally followed by a lightweight beautification pass. This division makes the enlarged MEC action space practical: MEC can route each request to the operation that best preserves what the user cares about, instead of forcing all visual tasks through a single generative or editing model.

Tool execution is constrained by the same executive role assigned to MEC. Consecutive read-only operations, such as web search, web fetch, image inspection, and memory retrieval, can be executed concurrently to reduce latency. Mutating or GPU-heavy operations, such as image generation, editing, composition, cropping, and text rendering, are serialized to preserve memory consistency and avoid resource contention. The loop tracks candidate images produced during a turn and can use an image-selection tool to choose the best final deliverable, keeping intermediate drafts from being over-exposed to the user. To prevent tool use from degenerating into uncontrolled loops, CMA-Harness adds simple controller-level constraints. If MEC repeatedly issues the same tool call with the same arguments, the system prompts it to revise its plan instead of executing the duplicate call indefinitely. Each user turn also has a maximum number of tool rounds; when this budget is exhausted, MEC summarizes the current progress and returns a graceful partial response. These mechanisms keep tool invocation under MEC control while still allowing multi-step workflows.

4.3. Flexible Persistent Multimodal Memory

CMA-Harness extends EVM from a session-local visual memory bank into a flexible persistent memory system. The Cognitive-structured Multimodal Agent focuses on controlled memory construction and retrieval within a fixed dialogue, whereas deployment requires memory to support multiple users, multiple sessions, reusable visual references, long transcripts, generated variants, fetched assets, and user feedback. We therefore organize memory into several scopes: session memory for the current interaction, user-level memory for durable preferences and project facts, gallery memory for reusable visual assets, and compact summaries for long-running conversations.

At the visual level, each image is stored as a three-part asset. The original image is preserved for future editing, composition, inspection, and download; a compact thumbnail is used for efficient retrieval and UI preview; and a JSON image card stores semantic tags, a short description, color palette, aspect ratio, file paths, and lifecycle metadata. The lifecycle fields are important in deployed workflows: parent-child links track edited variants, an “is current” flag records the active deliverable, user feedback marks preferred or rejected outputs, and selected images can be promoted into a reusable gallery. This turns PAE’s output from a tran-

sient caption into a structured memory object that can be updated and reused across turns.

Beyond visual assets, CMA-Harness maintains user-scope textual memories. These memories are stored as typed Markdown entries, including user preferences, feedback, project facts, and external references, with an automatically refreshed memory index. At prompt construction time, the memory manager assembles session summaries, user memories, and the visual memory index into a compact context block for MEC. This allows MEC to condition tool selection and response generation not only on the current dialogue, but also on persistent user intent and project-level constraints.

Retrieval is adaptive rather than uniform. Obvious references such as “the latest image”, “the first image”, or explicit image indices are resolved by deterministic rules without invoking a large model. When many candidates exist, a lightweight text prefilter narrows the candidate set using tags, descriptions, palette, recency, and feedback. The final selection uses multimodal retrieval over both image-card metadata and thumbnails, allowing CoRE to disambiguate references that are difficult to capture textually, such as “the blue one” or “the busier layout”. Full-resolution pixels are loaded only after a target image is selected and a downstream tool requires them.

The memory layer is also maintained asynchronously so that memory quality can improve without blocking the foreground interaction. When a new image is uploaded, generated, edited, fetched, or composed, CMA-Harness immediately stores the original image and thumbnail and can return the result to the user at once. A background extractor then analyzes the thumbnail and enriches the corresponding image card with semantic tags, a concise description, and color-palette information. This separates latency-sensitive interaction from slower memory annotation: the user does not wait for full visual abstraction, but subsequent turns can still benefit from richer retrieval cues.

Long conversations are handled similarly. Rather than replaying an unbounded transcript into MEC, CMA-Harness compacts older turns into structured summaries once a session exceeds token or tool-call thresholds, while keeping the most recent turns verbatim. The summary preserves high-level goals, established preferences, generated image identifiers, open follow-ups, and the current active state. Thus, long-term context remains available to the controller without continuously increasing the prompt size. Together, these mechanisms make EVM in CMA-Harness more flexible than the benchmark memory bank: it is multi-scope, multi-type, lifecycle-aware, and persistent, yet still preserves the token-efficient retrieval principle of the original Cognitive-structured Multimodal Agent.

4.4. Interactive Tool-Augmented Agent Execution

Finally, CMA-Harness turns the cognitive pipeline into an interactive execution loop. For each user turn, the system reconstructs the current memory context, rebuilds the system prompt with the latest memory blocks and tool schemas, lets MEC reason over the task, and executes any selected tools. Tool results are appended back into the transcript and then fed to MEC for the next decision, so the controller can iteratively search, retrieve, inspect, generate, edit, or compose until it has enough information to answer. This event-driven design keeps the foreground interaction responsive: intermediate reasoning, tool steps, tool results, images, and final text can be surfaced progressively while the same underlying MEC loop remains responsible for the complete decision process.

The deployed instance uses Qwen3.5-122B-A10B as the MEC planner, Qwen-Image-2512 for text-to-image generation, and Qwen-Image-Edit-2511 for editing and multi-image composition. These model choices are implementation instantiations of the modules rather than changes to the architecture. To make the agent practical for multi-step use, the planner and image backbones are kept as persistent components instead of being reloaded for every request. The generation and editing paths also include automatic prompt engineering: generation prompts are expanded toward richer visual descriptions, while editing prompts emphasize locality, object grounding, and preservation of unrelated regions.

The interaction loop also preserves the agent’s stateful multimodal semantics across turns. Uploaded or produced images are stored as addressable memory assets, tool outputs are recorded as part of the dialogue state, and the current deliverable is tracked so follow-up instructions such as “make it brighter” or “use the previous poster” can be resolved without reattaching every historical image. Lightweight safeguards, including duplicate-call detection, per-turn tool-round limits, and graceful partial summaries, keep the loop bounded while still allowing long multi-step workflows. Together, these choices turn the original Cognitive-structured Multimodal Agent design into a practical CMA-Harness: the same PAE–EVM–CoRE–MEC structure, but with a broader MEC action space, stronger persistent memory, and interactive execution for open-ended multimodal workflows.

5. Experiments

Our goal is to evaluate whether the proposed Cognitive-structured Multimodal Agent can effectively utilize long-horizon visual context while maintaining strong generation capability. To this end, we first describe the model setup (Sec. 5.1) and evaluation criteria (Sec. 5.2). We then compare our agent with unified and modular baselines on M2CA-Bench to measure context usage accuracy in multi-turn multimodal dialogues (Sec. 5.3) and analyze

how improved retrieval translates into downstream generation quality (Sec. 5.4). Finally, we examine the contribution of SFT/RL training (Sec. 5.5), compare against concurrent agent frameworks (Sec. 5.6), and ablate memory representations (Sec. 5.7).

5.1. Experimental Setup

Our system consists of three specialized vision-language modules (PAE, CoRE, and MEC) together with EVM. PAE performs visual abstraction by summarizing images into tags, description and thumbnails stored in the memory bank. CoRE conducts reasoning over the dialogue context and retrieves the relevant images from the memory bank. Finally, MEC integrates the retrieved visual context with the current dialogue state to perform task planning and generation.

Unless otherwise specified, all three modules are instantiated using Qwen3-VL-8B [4] as the backbone model. Image generation and editing are performed using Qwen-Image-Edit [37].

We compare our approach with three representative baselines:

- **Unified Model.** A unified multimodal model (BAGEL [7]) that directly handles the entire multi-turn dialogue within a single model.
- **Agent Baseline-8B/32B (All-Context).** A direct prompting baseline where the full conversational history (all preceding images and texts) is concatenated and fed into Qwen3-VL-8B/32B-Instruct for multimodal comprehension. Subsequently, Qwen-Image-Edit takes the comprehension output from Qwen3-VL as the conditional prompt, alongside all historical images, to execute the final generation. Since this end-to-end paradigm lacks an explicit retrieval mechanism, it inherently cannot be evaluated on our retrieval metrics. To address this, strictly during the retrieval evaluation phase, we explicitly prompt Qwen3-VL to output the indices of the target images. This intermediate index-prediction step is omitted during the actual generation experiments.
- **Multi-Agent Baseline.** A modular agent architecture where PAE, CoRE, and MEC are instantiated with Qwen3-VL-8B/32B-Instruct but trained without our long-horizon multimodal dialogue optimization.
- **Ours.** Our full agent trained on long-horizon multimodal multi-turn dialogues generated by the Unified Scenario Engine. The system is optimized through supervised fine-tuning (SFT) and reinforcement learning (RL) to improve visual abstraction and retrieval accuracy.

5.2. Evaluation Protocol and Metrics

We evaluate all methods on the proposed M2CA-Bench benchmark, which contains structured multi-turn multimodal dialogues with explicit image retrieval annotations.

We follow the benchmark construction and difficulty definition in Table 1.

Evaluation subsets. For each method we report results on three evaluation subsets:

- **Full:** performance across all 20 dialogue turns
- **Medium:** turns 11–20, emphasizing longer context
- **Hard:** the `very_hard` subset within turns 11–20

These settings simulate increasingly challenging long-horizon memory retrieval scenarios and allows us to measure how performance degrades as retrieval distance increases.

Metrics. We evaluate systems from two complementary perspectives:

- **Context Usage Accuracy.** Turn-level exact-match accuracy of the retrieved image index.
- **Generation Quality.** Per-turn generation quality scored from 0–10 using Gemini-3-Pro, which evaluates the faithfulness and visual correctness of the generated outputs.

Evaluation fairness. All methods are evaluated on identical scenario sessions, turn annotations, retrieval targets, and post-processing rules. Only the context utilization mechanism differs across methods, ensuring that performance differences are attributable to context modeling rather than pipeline implementation details.

5.3. Context Usage Accuracy

Table 3 reports turn-level retrieval accuracy. Several key observations emerge.

Unified models struggle with long-horizon retrieval. BAGEL, which lacks an explicit memory mechanism, achieves only 23.3% (EN) and 17.8% (CN) on Full, and its performance collapses to below 3% on the Hard subset. This indicates that unified architectures have difficulty isolating relevant visual context once the dialogue history becomes long.

Structured memory filtering improves context selection. The Multi-Agent baseline consistently outperforms the All-Context baseline. Using the 8B VLM, explicit memory filtering improves Full accuracy by +2.9% (EN) and +5.8% (CN), with larger gains on the harder Medium subset (+8.2%/+11.6%). This demonstrates that filtering irrelevant visual memories is essential in long multimodal conversations.

Task-specific training is more effective than parameter scaling. Increasing the Multi-Agent baseline from 8B to 32B yields only modest gains (+1.4% EN and +2.0% CN

on Full). In contrast, our 8B model, optimized with SFT and RL, surpasses the 32B baseline by +8.2% (EN) and +5.1% (CN). This suggests that targeted multimodal retrieval training is more impactful than increasing model capacity.

Advantages increase with task difficulty. The performance gap between our method and the 8B Multi-Agent baseline widens as retrieval difficulty increases: from +9.6% on Full, to +14.2% on Medium, and up to +18.4% on Hard. Moreover, our performance drop between Full and Hard is significantly smaller (9.4% vs. 18.2%), demonstrating stronger robustness in long-horizon visual recall.

5.4. Generation Quality

Table 4 reports generation quality scores evaluated by Gemini-3-Pro.

Generation quality closely follows retrieval accuracy. Our method achieves the highest overall scores (8.49 EN / 8.53 CN). The ranking of methods closely mirrors the retrieval accuracy results, indicating that correct visual context retrieval remains the primary bottleneck in long-horizon multimodal generation.

Task-level analysis. **Generate** tasks are least sensitive to retrieval ($R^* = \emptyset$), and most agent-based methods achieve high scores (above 7.1). **Edit** tasks are the most challenging, typically scoring between 6.1 and 7.5, since they require both accurate source retrieval and faithful instruction adherence. Our method achieves the highest edit score of 7.44 (CN) while remaining competitive with 7.31 (EN). **Understand/Chat** tasks benefit most from improved retrieval. Our method reaches 8.87 in English, outperforming the Multi-Agent Baseline-32B by +0.65.

Retrieval as a quality multiplier. Comparing the Agent Baseline-8B (All-Context) with our method shows that a roughly 10% gain in retrieval accuracy translates into generation quality improvements of +0.60 (EN) and +1.43 (CN). This demonstrates a strong multiplier effect, where improvements in context selection propagate through the entire generation pipeline.

5.5. Training Stage Analysis

To understand the contribution of different training stages to the agent’s performance, we analyze the sequential optimization of the Perceptual Abstraction Engine (PAE) and Cognitive Retrieval Engine (CoRE), which involve both supervised fine-tuning (SFT) and reinforcement learning (RL).

Our training pipeline consists of three stages. Table 5 reports the retrieval accuracy on the challenging Hard subset after each stage:

Table 3. Pipeline retrieval results on M2CA-Bench.

Method	English			Chinese		
	Full	Medium	Hard	Full	Medium	Hard
Unified Model (BAGEL)	23.3%	5.6%	1.9%	17.8%	3.2%	2.3%
Agent Baseline-8B (All-Context)	78.9%	67.0%	59.5%	76.7%	66.0%	60.8%
Agent Baseline-32B (All-Context)	81.9%	72.1%	62.0%	80.3%	72.5%	71.4%
Multi-Agent Baseline-8B	81.8%	75.2%	63.6%	82.5%	77.6%	77.8%
Multi-Agent Baseline-32B	83.2%	79.4%	72.1%	84.5%	80.0%	79.0%
Ours (8B)	91.4%	89.4%	82.0%	89.6%	85.6%	79.0%



Figure 4. Qualitative comparison between the baseline and our method on multi-turn visual understanding, generation, and editing.

- **Stage 1 – SFT on CoRE:** Applying supervised fine-tuning to the retrieval module improves Hard accuracy by 7.3%, indicating that the model can better extract relevant visual memories from the structured dialogue data.
- **Stage 2 – RL on CoRE:** Reinforcement learning further refines retrieval strategies, yielding an additional 6.5% improvement and demonstrating that the agent can optimize long-horizon memory selection beyond supervised signals.
- **Stage 3 – RL on PAE:** Finally, RL applied to the perception module enhances visual abstraction and memory encoding, adding 4.6% more accuracy. This stage ensures that the most informative representations are stored for downstream retrieval and generation.

These results confirm that a carefully staged combination of SFT and RL effectively unlocks the agent’s ability to

utilize multimodal context over long dialogues.

We further assess the computational efficiency of our decoupled architecture. Table 6 compares our approach with the Agent Baseline-32B, which processes the entire multimodal history in a single model. By delegating memory filtering to PAE and CoRE, the agent retrieves only relevant visual memories instead of the full context. This targeted retrieval reduces the number of visual tokens fed into the generation module, improving efficiency: inference time per turn is nearly halved (12.7s vs. 23.1s), while generation quality also increases (8.77 vs. 7.93).

Qualitative examples in Figure 4 further illustrate the benefits of staged training and modular retrieval. In the all-context baseline, lack of filtering causes hallucination: the agent over-relies on the first image and incorrectly propagates its content into subsequent generations. In contrast,

Table 4. **Gemini generation quality on M2CA-Bench.** Image Generation and Understanding/Chat are abbreviated as Gen. and U./C., respectively.

Method	English				Chinese			
	Overall	Gen.	Edit	U./C.	Overall	Gen.	Edit	U./C.
Unified Model (BAGEL)	4.97	5.11	4.09	5.77	4.77	5.68	3.39	4.97
Agent Baseline-8B (All-Context)	7.89	8.85	6.71	8.05	7.10	7.86	6.13	7.15
Agent Baseline-32B (All-Context)	7.93	9.14	6.38	8.19	7.45	8.31	6.25	7.53
Multi-Agent Baseline-8B	8.07	9.04	7.33	7.75	7.63	8.35	6.19	8.03
Multi-Agent Baseline-32B	8.24	8.97	7.49	8.22	7.91	9.24	7.03	7.34
Ours (8B)	8.49	9.26	7.31	8.87	8.53	9.38	7.44	8.56

Table 5. Ablation Study on the Training Process for Memory Construction and Retrieval.

Method	Full	Medium	Hard
Multi-Agent Baseline-8B	81.8%	75.2%	63.6%
+ SFT (CoRE)	86.6%	83.7%	70.9%
+ RL (CoRE)	90.1%	88.3%	77.4%
+ RL (PAE)	91.4%	89.4%	82.0%

Table 6. Ablation study showing the efficiency and quality gains from the proposed PAE +CoRE modules.

Method	Full	Hard	Quality	Runtime (s)
Agent-Baseline-32B	81.9%	62.0%	7.93	23.1
PAE +CoRE + Qwen3VL-32B	92.0%	83.5%	8.77	12.7

Table 7. Multi-turn generation and editing quality scores (0–10) on English M2CA-Bench. Sessions are split into early and late stages to highlight robustness against long-horizon semantic drift.

Method	Generation Quality		Editing Quality	
	Early	Late	Early	Late
GenArtist	6.77	4.80	5.29	4.03
Ours (8B Agent)	9.51	9.06	7.74	7.13

our staged training ensures accurate retrieval and, coupled with MEC’s prompt rewriting, produces outputs that adhere closely to user instructions across multi-turn interactions.

5.6. Comparison with Concurrent Agent Frameworks

We further compare our agent with GenArtist [36], a concurrent multimodal agent framework for unified image generation and editing. To expose long-horizon drift, we split English M2CA-Bench sessions into early (turns 1–10) and late (turns 11–20) stages and separately evaluate generation and editing quality. Table 7 shows that GenArtist degrades substantially in late turns, while our agent remains stable. This result supports the central claim that tool orchestration alone is insufficient: without explicit episodic visual memory and retrieval, multi-turn generation and editing suffer from context confusion as sessions grow.

Table 8. Ablation on M2CA-Bench comparing text-only LLM memory systems against our full EVM. Retrieval accuracy (%) is reported across difficulty subsets.

Memory Strategy	Full	Medium	Hard
MemoryLLM-style memory	51.7%	47.2%	24.4%
ReasoningBank-style memory	68.2%	51.8%	39.0%
Ours (Full EVM)	91.4%	89.4%	82.0%

5.7. Ablation on Memory Representations

Text-only LLM memories provide useful long-term semantic state, but they are not sufficient for fine-grained visual recall. We evaluate MemoryLLM-style and ReasoningBank-style text memories by supplying agents with textual abstractions of images and asking them to retrieve the relevant historical visual episode. As shown in Table 8, both text-only strategies degrade sharply on the Hard subset, where queries often require cross-topic retrieval, long-range callbacks, or subtle visual discrimination. In contrast, our full EVM retains compressed thumbnails together with textual metadata and achieves substantially higher accuracy.

6. Qualitative Examples

We include additional long-horizon dialogue examples to illustrate how episodic visual memory supports mixed generation, editing, composition, and visual question answering. One example begins with a Mars habitat, later edits it with a dust storm, and then correctly answers questions about earlier visual details such as solar panels. A second example performs environmental transformations such as converting a sunny street scene into a rainy night version while preserving spatial layout, then recalls distant details from a separate savanna scene. A third example spans indoor scenes, logistics environments, wildlife landscapes, and Mars habitats, requiring object insertion, cross-scene composition, and detailed visual comparison. Across these examples, the agent demonstrates long-range visual recall, topic switching without memory loss, multi-image composition, and grounded multimodal reasoning.

Additional qualitative example 1 placeholder: case1.pdf is not included in the source tree.

Figure 5. **Example Dialogue 1.** A long-horizon multimodal interaction containing image generation, editing, topic switching, and visual question answering.

Additional qualitative example 2 placeholder: case2.pdf is not included in the source tree.

Figure 6. **Example Dialogue 2.** A dialogue emphasizing environmental transformations, long-range recall, and stable visual consistency.

7. Conclusion

We propose a cognitive-structured multimodal agent that externalizes visual history into episodic memory and selectively retrieves relevant episodes, enabling effective long-horizon multimodal dialogue. Together with the M2CA-Bench benchmark and a staged SFT+RL training pipeline, our 8B agent achieves 91.4% retrieval accuracy over 20-turn sessions, surpassing 32B baselines by +8.2% while nearly halving inference time. We further show that the same memory-centric decomposition can be instantiated as CMA-Harness, a deployed tool-augmented multimodal assistant with persistent visual memory, web access, image generation/editing/composition, and API serving. This trajectory connects controlled benchmark optimization with a practical open-ended agent system. Future directions include tighter integration between the trained retrieval modules and the deployed tool stack, end-to-end joint optimization, open-domain extension, and richer modality support.

Additional qualitative example 3 placeholder: case3.pdf is not included in the source tree.

Figure 7. **Example Dialogue 3.** A complex interaction involving object insertion, cross-scene composition, and detailed visual comparison across distant turns.

A. Prompt Templates

We provide the core prompt templates used by the agent modules to clarify the module interfaces and support reproducibility. The wording is intentionally simple and format-constrained, so that memory writing, retrieval, task classification, prompt rewriting, and visual answering expose stable input-output contracts across training, evaluation, and deployment.

PAE Prompt (Image → Memory)

```
<image> Please analyze this image in detail and generate complete image features + description in English. The output must be JSON containing the following fields: tags (tag list, at least 5 tags, array format), description (detailed description, at least 100 words). Please ensure the output is valid JSON.
```

CoRE Prompt (Retrieval)

```
Candidate image list (aligned with the order in 'images'):  
- rank 0: <image>  
- rank 1: <image>  
...  
Memory bank information (including tags + description):  
{[0] | tags: sunset, sports car, highway  
| description: A sleek black ...}  
{[1] | tags: beach, ocean, sunset |  
description: A panoramic view ...}  
...  
User query: {q}  
Please select the most relevant image ID list based on the user query and memory bank information. If no historical image is needed, output an empty list.
```

MEC Prompt (Task Classification)

```
You are a multimodal task classification expert. Given the user input and current state, categorize the task into one of the following five types: generate, edit, composite, understand, pure.chat.  
Current state: User text = {user.input};  
Image status = {image.info}.  
Output requirement: Return JSON only, with no explanation: {"task.type": "generate" | "edit" | "composite" | "understand" | "pure.chat"}.
```

MEC Prompt (Generate Mode Rewriting)

```
You are a prompt rewriting assistant for text-to-image generation models. Convert a short user instruction into a detailed visual prompt. Identify the main subject, expand visual attributes such as color, shape, size, and appearance, add reasonable scene or environment details if not specified, optionally include viewpoint or composition, and keep the user's original intent unchanged. Output only the final rewritten prompt. User instruction: {instruction}.
```

MEC Prompt (Edit Mode Rewriting)

```
You are a multimodal prompt rewriting assistant for image editing models. You will receive an image, a user editing instruction, and optional dialogue context. Identify the object or region to modify using the image, clearly describe the requested modification, preserve the user's intent, explicitly indicate that all other parts of the image should remain unchanged, and output only one concise editing prompt. Image editing instruction: {instruction}.
```

MEC Prompt (Understanding / Chat)

```
Text-only mode: Current user instruction = {user.instruction}; task type = pure.chat. Please answer directly.  
Contextual mode: <image> <image> ...; dialogue history; retrieved memories with tags and descriptions; current user instruction = {user.instruction}; task type = mmu. Please reason carefully based on the instruction and reference images, and provide a detailed answer.
```

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023.
- [3] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou.

- Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. 2023.
- [4] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.
 - [5] Tim Brooks, Aleksander Holynski, and Alexei A Efros. Instructpix2pix: Learning to follow image editing instructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18392–18402, 2023.
 - [6] Chameleon Team. Chameleon: Mixed-modal early-fusion foundation models. *arXiv preprint arXiv:2405.09818*, 2024.
 - [7] Chaorui Deng et al. Emerging properties in unified multimodal pretraining. *arXiv preprint arXiv:2505.14683*, 2025.
 - [8] Yue Fan, Xiaojian Ma, Rujie Wu, Yuntao Du, Jiaqi Li, Zhi Gao, and Qing Li. Videoagent: A memory-augmented multimodal agent for video understanding. In *European Conference on Computer Vision*, pages 75–92. Springer, 2024.
 - [9] Zhi Gao, Bofei Zhang, Pengxiang Li, Xiaojian Ma, Tao Yuan, Yue Fan, Yuwei Wu, Yunde Jia, Song-Chun Zhu, and Qing Li. Multi-modal agent tuning: Building a vlm-driven agent for efficient tool usage. *arXiv preprint arXiv:2412.15606*, 2024.
 - [10] Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xianwu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, et al. Metagpt: Meta programming for a multi-agent collaborative framework. In *The twelfth international conference on learning representations*, 2023.
 - [11] Yujie Hu et al. Talkphoto: A versatile training-free conversational assistant for intelligent image editing. *arXiv preprint arXiv:2601.01915*, 2026.
 - [12] Zhipeng Huang, Shaobin Zhuang, Canmiao Fu, Binxin Yang, Ying Zhang, Chong Sun, Zhizheng Zhang, Yali Wang, Chen Li, and Zheng-Jun Zha. Wegen: A unified model for interactive multimodal generation as we chat. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 23679–23689, 2025.
 - [13] Hanqi Jiang et al. Synapse: Empowering llm agents with episodic-semantic memory via spreading activation. *arXiv preprint arXiv:2601.02744*, 2026.
 - [14] Hongbo Jin, Qingyuan Wang, Wenhao Zhang, Yang Liu, and Sijie Cheng. Videomem: Enhancing ultra-long video understanding via adaptive memory management. *arXiv preprint arXiv:2512.04540*, 2025.
 - [15] Somnath Kumar, Yash Gadhia, Tanuja Ganu, and Akshay Nambi. Mmctagent: Multi-modal critical thinking agent framework for complex visual reasoning. *arXiv preprint arXiv:2405.18358*, 2024.
 - [16] Guohao Li, Hasan Abed Al Abbas Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: Communicative agents for “mind” exploration of large language model society. In *Advances in Neural Information Processing Systems*, 2023.
 - [17] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR, 2023.
 - [18] Pengxiang Li et al. Iterative trajectory exploration for multimodal agents. In *Advances in Neural Information Processing Systems*, 2025.
 - [19] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, 2024.
 - [20] Jiaqi Liu, Kaiwen Xiong, Peng Xia, Yiyang Zhou, Haonian Ji, Lu Feng, Siwei Han, Mingyu Ding, and Huaxiu Yao. Agent0-vl: Exploring self-evolving agent for tool-integrated vision-language reasoning. *arXiv preprint arXiv:2511.19900*, 2025.
 - [21] Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory. *arXiv preprint arXiv:2508.09736*, 2025.
 - [22] Yiyang Ma et al. Janusflow: Harmonizing autoregression and rectified flow for unified multimodal understanding and generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
 - [23] Siru Ouyang, Jun Yan, I Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T Le, Samira Daruki, Xiangru Tang, et al. Reasoningbank: Scaling agent self-evolving with reasoning memory. *arXiv preprint arXiv:2509.25140*, 2025.
 - [24] Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shaohan Huang, Shuming Ma, and Furu Wei. Kosmos-2: Grounding multimodal large language models to the world. *arXiv preprint arXiv:2306.14824*, 2023.
 - [25] Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, et al. Chatdev: Communicative agents for software development. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)*, pages 15174–15186, 2024.
 - [26] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36, 2024.
 - [27] Hao Tang, Chenwei Xie, Xiaoyi Bao, Tingyu Weng, Pandeng Li, Yun Zheng, and Liwei Wang. Unilip: Adapting clip for unified multimodal understanding, generation and editing. *arXiv preprint arXiv:2507.23278*, 2025.
 - [28] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
 - [29] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
 - [30] Kavana Venkatesh et al. Crea: A collaborative multi-agent framework for creative content generation with diffusion models. In *Advances in Neural Information Processing Systems*, 2025.
 - [31] Guoyan Wang, Yanyan Huang, Chunlin Chen, Lifeng Wang, and Yuxiang Sun. Yanyun-3: Enabling cross-platform strategy

- game operation with vision-language models. *arXiv preprint arXiv:2511.12937*, 2025.
- [32] Hengyi Wang et al. Multimodal needle in a haystack. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2025.
- [33] Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.
- [34] Xinlong Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Yufeng Cui, Jinsheng Wang, Fan Zhang, Yueze Wang, Zhen Li, Qiying Yu, et al. Emu3: Next-token prediction is all you need. *arXiv preprint arXiv:2409.18869*, 2024.
- [35] Xiaohan Wang, Yuhui Zhang, Orr Zohar, and Serena Yeung-Levy. Videoagent: Long-form video understanding with large language model as agent. In *European Conference on Computer Vision*, pages 58–76. Springer, 2024.
- [36] Zhenyu Wang, Aoxue Li, Zhenguo Li, and Xihui Liu. Genartist: Multimodal llm as an agent for unified image generation and editing. *Advances in Neural Information Processing Systems*, 37:128374–128395, 2024.
- [37] Chenfei Wu, Jiahao Li, Jingren Zhou, Junyang Lin, Kaiyuan Gao, Kun Yan, Sheng ming Yin, Shuai Bai, Xiao Xu, Yilei Chen, Yuxiang Chen, Zecheng Tang, Zekai Zhang, Zhengyi Wang, An Yang, Bowen Yu, Chen Cheng, Dayiheng Liu, Deqing Li, Hang Zhang, Hao Meng, Hu Wei, Jingyuan Ni, Kai Chen, Kuan Cao, Liang Peng, Lin Qu, Minggang Wu, Peng Wang, Shuting Yu, Tingkun Wen, Wensen Feng, Xiaoxiao Xu, Yi Wang, Yichang Zhang, Yongqiang Zhu, Yujia Wu, Yuxuan Cai, and Zenan Liu. Qwen-image technical report, 2025.
- [38] Tsung-Han Wu et al. Visual haystacks: A vision-centric needle-in-a-haystack benchmark. *arXiv preprint arXiv:2407.13766*, 2024.
- [39] Jinheng Xie et al. Show-o2: Improved native unified multimodal models. *arXiv preprint arXiv:2506.15564*, 2025.
- [40] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghai Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [41] Rui Ye, Zhongwang Zhang, Kuan Li, Huifeng Yin, Zhengwei Tao, Yida Zhao, Liangcai Su, Liwen Zhang, Zile Qiao, Xinyu Wang, et al. Agentfold: Long-horizon web agents with proactive context management. *arXiv preprint arXiv:2510.24699*, 2025.
- [42] Woongyeong Yeo et al. Worldmm: Dynamic multimodal memory agent for long video reasoning. *arXiv preprint arXiv:2512.02425*, 2025.
- [43] Qiying Yu et al. Dapo: An open-source llm reinforcement learning system. *arXiv preprint arXiv:2503.14476*, 2025.
- [44] Zijun Zhou et al. Multi-turn consistent image editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.
- [45] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*, 2023.